RAJIV GANDHI INSTITUTE OF TECHNOLOGY

GOVERNMENT ENGINEERING COLLEGE

KOTTAYAM-686 501

DEPARTMENT OF
COMPUTER SCIENCE AND ENGINEERING

CSD 334 MINI PROJECT

INTERNAL EXAM MANAGEMENT SYSTEM

Report Submitted by

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APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY
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CERTIFICATE

This is to certify that this report entitled ***INTERNAL EXAM MANAGEMENT SYSTEM*** *is an authentic report of the project done by* ***Amal Krishna K P (KTE22CS013), Muhammad Shuhaibh (KTE22CS048), Muhammed Adhil P P (KTE22CS049) and Nadha K M (KTE22CS050)*** *during the academic year* ***2024-25****, in partial fulfillment of the requirements for the award of the Degree of Bachelor of Technology in Computer Science and Engineering of APJ Abdul Kalam Technological University, Thiruvananthapuram.*

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Declaration

We, the undersigned hereby declare that the project report entitled **Internal Examination Management System**, submitted for partial fulfilment of the requirements for the award of the degree of Bachelor of Technology of the **APJ Abdul Kalam Technological University, Kerala** is a bonafide work done by us under the supervision of **Prof. Anil Kumar S**. This submission represents our idea in our own words where ideas or words of others have not been included; we have adequately and accurately cited and referenced the original sources. We have adhered to the ethics of academic honesty and integrity and have not misrepresented or fabricated any data, idea or on our submission. We understand that any violation of the above can result in disciplinary actions from the institute and/ or University and can evoke penal action from the sources which have not been properly cited or from whom proper permission has not been obtained. This report has not been previously formed as the basis for awarding any degree, diploma or similar title of any other university.

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Abstract

The *Internal Exam Management System* is a comprehensive web application designed to streamline various aspects of conducting internal exams in educational institutions. Key functionalities include enabling teachers to upload draft versions of question papers, facilitating review and approval workflows for exam coordinators, and implementing automated student seating arrangements.

This project addresses common challenges associated with manual exam management systems, such as disorganized paper submissions, inefficient approval processes, and the complexities of creating optimal seating arrangements. By providing a centralized and secure platform, the system simplifies communication between teachers and coordinators while ensuring efficient allocation of seating based on predefined criteria.

With a user-friendly interface, the software reduces manual effort, minimizes errors, and enhances coordination. It significantly improves the quality and organization of internal exams by automating essential tasks and promoting effective collaboration among all stakeholders.

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Chapter 1

Introduction

Manual internal exam management in the CSE Department at RIT Kottayam results in disorganized question paper submissions, inefficient scrutiny, and complex seating arrangements, leading to increased workload, errors, and miscommunication. The lack of an automated system makes coordination among faculty members and examination coordinators challenging. The Internal Examination Management System provides a centralized web-based solution to streamline question paper handling, automate seating arrangements, and enhance overall exam organization. By reducing manual effort and minimizing errors, the system improves efficiency, accuracy, and coordination, ensuring a structured and hassle-free examination process.

1.1 Objectives

This project aims to achieve the following objectives:

1. To enable faculty members to upload question papers, facilitate scrutiny by coordinators, and ensure a structured approval workflow.

2. To implement an automated system to allocate students to examination halls based on predefined criteria, reducing manual effort and errors.

3. To provide a centralized platform for seamless interaction among Examination Coordinators, Module Coordinators, and faculty members.

4. To minimize errors in examination management by reducing manual interventions and implementing automated processes.

5. To implement role-based access control to ensure that only authorized personnel can access and manage examination-related data.

1.2 Motivation

Effective exam management ensures fairness, accuracy, and smooth administration. The manual system in the CSE Department at RIT Kottayam is inefficient, error-prone, and time-consuming, highlighting the need for automation. Key motivations for developing this system include:

1. *Eliminating Manual Errors:-* Reducing mistakes in question paper handling, scrutiny, and seating arrangements.
2. *Enhancing Efficiency:-* Automating repetitive tasks to save time and effort for faculty and coordinators.
3. *Improving Coordination:-* Providing a centralized platform for seamless communication and structured exam management.

1.3 Scope of the Project

1. *Need:-*
   - Lack of a proper automated system for handling Internal Examination in the Computer Science and Engineering Department of RIT Kottayam.
2. *Deliverables:-*
   - A web application that enables to upload question papers, conduct question paper scrutiniy, set seating arrangement and allocate invigilation duty to the faculty.
3. *Assumptions:-*
   - Users have the basic knowledge of handling web-based applications.
   - Users have clear idea on examination management.

1.4 Prerequisites for the Reader

To understand the contents of this report, the reader is expected to meet the following prerequisites:

1. *Basic Knowledge of Examination Management:-* Familiarity with internal examination procedures, including question paper submission, scrutiny, and seating arrangements.
2. *Understanding of Web-Based Systems:-* Awareness of web applications and their role in automating administrative tasks.
3. *Technical Background:-* Basic knowledge of database management, user authentication, and role-based access control in software systems.

1.5 Organization of the Report

The remainder of the report is organized into six chapters. In Chapter 2, we detail the system study and the requirement engineering process. This includes the details of the existing systems (section 2.1), gap analysis (section 2.2), the proposed system (section 2.3), and those of the requirements engineering process (section 2.4), including feasibility study (section 2.4.1), requirements elicitation (section 2.4.2), requirements specification (section 2.4.4), and requirements validation (section 2.4.5). In Chapter 3, we detail the design of the proposed system, by giving a detailed discussion on system design (section 3.1) and the detailed design (section 3.2). Chapter 4 is intended to give the details of the implementation of the proposed system by providing an account of what tools are used for the implementation (section 4.1) and how each module is implemented using these tools (section 4.3). In Chapter 5, we discuss the different testing strategies used in this project (section 5.1) and the results of applying these strategies to the developed system (section 5.2). Chapter 6 is dedicated to presenting the results obtained (section 6.1) along with a result analysis (section 6.2). We close the report with chapter 7 which is a summary and recommendations for future works (section 7.1).

Chapter 2

System Study and Requirements Engineering

System Study is a critical phase in software development that involves analyzing existing systems, identifying requirements, and proposing improvements. It aims to understand the system's functionality, assess its limitations, and align it with user needs. It lays the foundation for successful software development, ensuring that the final product meets user expectations and business objectives.

2.1 Existing Systems

Currently, internal examination management in the CSE Department at RIT Kottayam relies on a manual process involving paperwork, spreadsheets, and email communication. Question paper submissions, scrutiny, seating arrangements, and coordination are handled manually, leading to inefficiencies and errors, which are time-consuming.

2.1.1 Literature Review

Literature Survey involves an extensive search and review of existing published literature, academic papers, journals, books, and relevant online sources related to the subject of study. The purpose of the literature survey is to gain a comprehensive understanding of the existing knowledge, theories, and findings on the topic under investigation. It is a critical component of academic research or project reports. It helps researchers identify gaps in the existing knowledge or areas that have not been extensively studied, providing opportunities for the new research and contributions. While no dedicated system is currently implemented in the department, various studies and solutions have addressed similar challenges in educational institutions. Key findings from related works include:

- Automated Examination Management Systems: Research highlights the benefits of digitizing exam-related processes, improving efficiency, accuracy, and security.

- Seating Arrangement Algorithms: Studies on optimization techniques for seating allocation demonstrate how automation can minimize conflicts and maximize space utilization.

- Question Paper Scrutiny Systems: Existing frameworks emphasize the importance of multi-level verification to maintain the integrity and quality of exam papers.

2.2 Gap Analysis

The existing manual exam management system is time-consuming, error-prone, and lacks coordination. Processes like question paper scrutiny, seating arrangements, and duty allocation rely on paperwork and emails, leading to inefficiencies and miscommunication. The proposed system automates these tasks, reducing errors, saving time, and improving coordination. It ensures secure data handling, centralized communication, and role-based access control, making exam management more efficient and reliable.

2.3 Proposed System

The Internal Examination Management System is designed to replace the manual examination management process in the CSE Department at RIT Kottayam with an automated, web-based solution. It simplifies question paper management, and seating arrangements while ensuring efficiency, accuracy, and security. The system enables seamless coordination among faculty members, Examination Coordinators, and the HOD, eliminating delays and errors caused by the existing manual system.

Advantages

- Automated Exam Management: Streamlines question paper submission, scrutiny,and seating arrangement, reducing manual effort.
- Reduced Errors and Inconsistencies: Eliminates manual data handling mistakes, ensuring accurate seating plans and schedules.
- Reduced Errors and Inconsistencies: Eliminates manual data handling mistakes, ensuring accurate seating plans and invigilation schedules.
- Better Coordination and Communication: A centralized platform ensures smooth interaction between faculty, coordinators, and HODs.
- Secure and Controlled Access: Role-based authentication restricts access to sensitive information, protecting question papers and seating data.
- User-Friendly Interface: Simple and intuitive design ensures that faculty and coordinators can navigate and use the system with ease.

Disadvantages

- Initial Learning Curve: Faculty and coordinators may require training to adapt to the new digital system.
- Dependence on Internet Connectivity: The system requires a stable internet connection for smooth operation.
- Development and Maintenance Costs: Although it uses open-source technologies, some server and infrastructure costs may be involved in deployment and maintenance.
- Potential Resistance to Change: Some faculty members may be hesitant to shift from traditional manual methods to an automated system.

2.3.1 Problem Statement

To develop a web-based internal examination management system that simplifies and automates various exam-related activities. The system will help reduce errors, improve efficiency, and ensure a smooth examination process.

2.3.2 System Model

The model of the proposed system is shown in Figure 2.1. The system consists of four main user groups:

1. Head of Department (HOD): Responsible for managing user accounts, modules, subject profiles, and batch and room details. HOD ensures that the basic configurations for the internal exam process are set up.
2. Exam Coordinator/Assistant Exam Coordinator (EC/AEC): Handles the allocation of invigilation duties and seating arrangements. These processes ensure proper exam administration. Notifications are generated as part of these tasks.
3. Module Coordinator(MC): Reviews question papers to ensure correctness and fairness. This process includes scrutiny, where remarks may be added, and notifications may be sent regarding issues.
4. Faculty: Uploads question papers and manages error handling related to them. If any discrepancies are found, additional actions such as error handling may be performed.

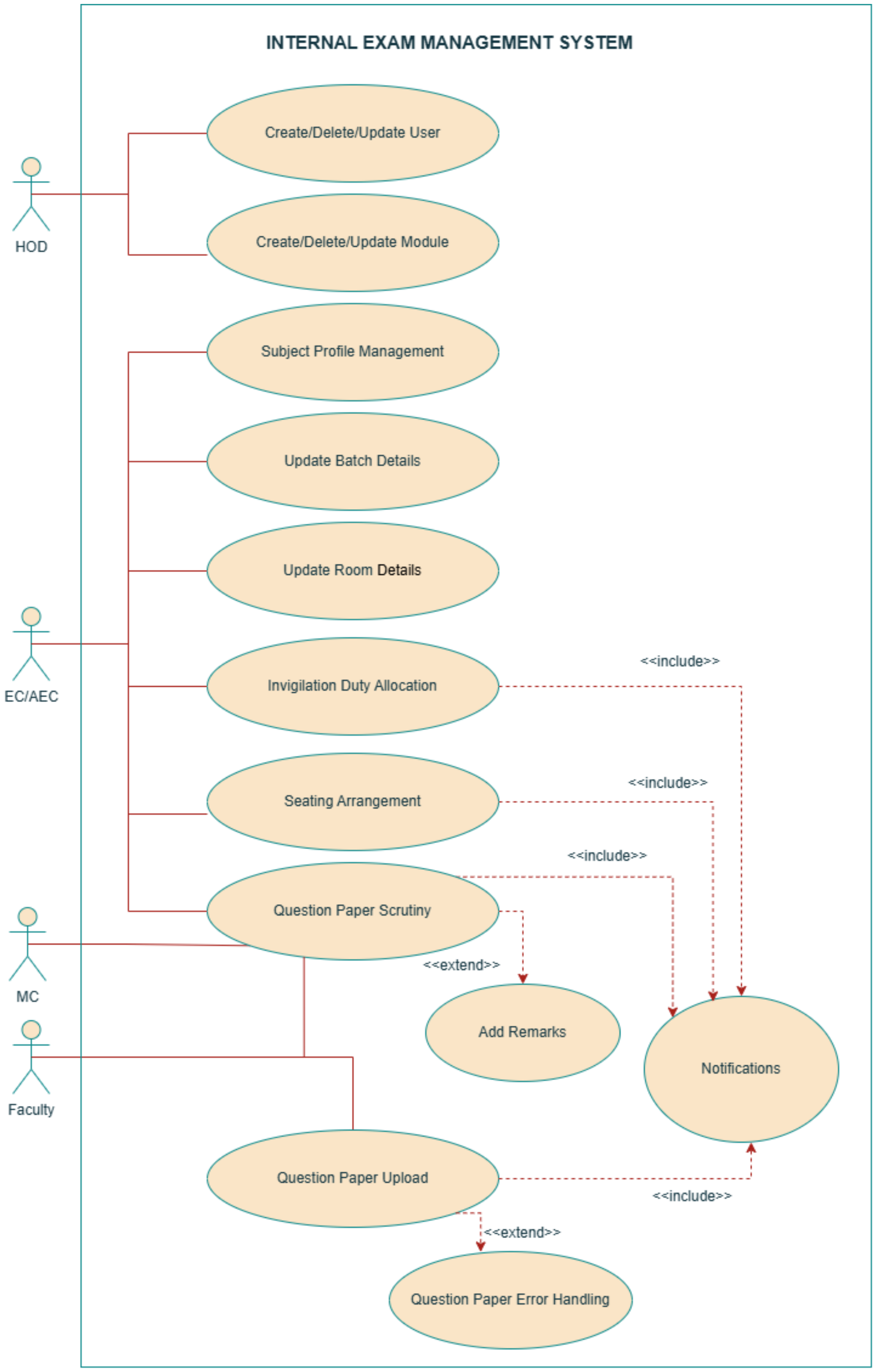

Figure 2.1: System Model

2.4 Requirements Engineering

The process of gathering software requirements from clients and then analysing and documenting them is known as requirements engineering. The goal of requirement engineering is to develop and maintain a sophisticated and descriptive 'System Requirements Specification' document. It essentially involves the following five activities.

1. Feasibility Study
2. Requirements Elicitation
3. Requirements Analysis
4. Requirements Specification
5. Requirements Validation

2.4.1 Feasibility Study

Feasibilty check [3,4] is a procedure that identifies, describes and evaluates the proposed systems and selects the best system for the task under consideration. An estimate is made of whether the identified user needs may be satisfied using current software and hardware technologies. The study will decide if the proposed system will be cost-effective from a business point of view and if it can develop given existing budgetary constraints. The key considerations involved in the feasibility analysis are

- Economic Feasibility
- Technical Feasibility
- Behavioural Feasibility

2.4.1.1 Economic Feasibility

Economic analysis [3,4] refers to whether a project is financially viable and makes sense from a cost perspective. It involves analyzing the potential costs and benefits to determine if the project can be completed within the available budget and if it will generate enough returns to justify the cost. If the returns outweigh the costs, the project is considered economically feasible, and it's more likely to be pursued. However, if the costs exceed the expected benefits or the financial riks are too high, the project may not be considered economically feasible, and other alternatives will be explored. For the proposed system, economic feasibility is ensured by using open-source technologies like Node.js and MongoDB, reducing software costs. Automating scrutiny, seating, and duty allocation lowers administrative workload and expenses. The system's time savings, error reduction, and improved coordination outweigh development costs, making it a cost-effective, web-based solution with minimal hardware requirements.

2.4.1.2 Technical Feasibility

Technical Feasibility [3,4] refers to whether a proposed project or system can be successfully developed, implemented, and integrated within the existing technological infrastructure. It invloves assessing the availability of required technology, expertise, and resources to accomplish the project's objectives. If the necessary technology and expertise are available, and any potential technical challenges can be overcome, the project is considered technically feasible; otherwise, it may face limitations or require significant modifications for successful implementation. The proposed system is technically feasible, using Node.js, MongoDB, HTML, CSS, and JavaScript, which are well-supported and scalable. It can be deployed on existing infrastructure with minimal hardware needs. With the required expertise available, integration and implementation are smooth and reliable.

2.4.1.3 Behavioural Feasibility

This is also known as *Operational Feasibilty*[3,4]. Behavioural Feasibility refers to the assessment of whether a proposed system or project is acceptable and embraced by its intended users and stakeholders. A system that aligns with user's needs, preferences, and existing behaviours is more likely to be considered behaviourally feasible, leading to higher user satisfaction and successful implementation. The system is behaviorally feasible as it simplifies exam management tasks for faculty and coordinators, reducing workload and errors. Its user-friendly interface ensures easy adoption with minimal training. By aligning with existing examination workflows, it enhances efficiency and is more likely to be accepted by users.

2.4.2 Requirements Elicitation

Requirement elicitation is the process of collecting or gathering and identifying user needs, expectations, and system functionalities to define the scope and specifications of a project or software system.

1. Interview: Formal or informal interviews with stakeholders and end users are part of most requirements engineering processes. In these interviews, the requirements engineering team ask questions to stakeholders about the system that they currently use and the system to be developed. The answers to these questions gives the proper requirements for our system. Discussions were held with the Examination Coordinator

and Assistant Examination Coordinator to understand the challenges in the current manual examination management process.

2. Study of Existing Systems: As mentioned earlier, since no dedicated system was in place, we analyzed the manual procedures followed for question paper scrutiny, seating arrangements, and duty allocation to identify areas for automation. Details are given in section 2.1.

3. Surveys: A survey was conducted among all faculty members in the department to collect feedback on pain points and expectations from the proposed system.

2.4.3 Requirements Analysis

Requirement analysis is the process of documenting, and evaluating user needs and expectations for a software system or project. It forms the basis for successful system design and development, ensuring that the final product meets stakeholders' objectives and fulfills user requirements. We analyzed the data collected from the interview and the study of existing systems and valuable information was extracted from these data. This information led to the development of the proposed system with the following requirements.

1. User Roles & Authentication: Secure login for HOD, Examination Coordinator, Assistant Examination

2. Question Paper Management: Upload, scrutiny, and approval workflow.

3. Seating Arrangement Automation: Automatic generation of seating plans.

4. Notifications & Updates: Instant alerts for important exam-related tasks.

2.4.4 Requirements Specification

When a requirements specification is written, it is important to remember that the main goal is to deliver the best product possible, and not to produce a perfect requirements specification. There are many good definitions that describe how a requirements specification should be written, but all have at least one part in common, which is the essence of requirements specification, namely the requirements. Requirements are divided into *functional requirements* and *non-functional requirements*[3,4]. Functional requirements describe the functionality of the desired system that usually consists of some kind of calculation and results, given specific inputs. Non-functional requirements[3,4] describe how quickly these calculations should be and how quickly the system will respond when its functionality is used. The SRS[3,4] document is given in the Appendix.

2.4.5 Requirements Validation

Requirement validation is the process of evaluating and verifying that the gathered requirements are accurate, complete, and consistent. It ensures that the documented requirements align with stakeholders' actual needs and that they can be realistically achieved within the project's constraints. The validation process ensures that the Internal Examination Management System effectively addresses manual exam management challenges while aligning with faculty and coordinator needs. Through discussions and surveys, we confirmed feasibility, refined requirements, and ensured smooth implementation.

2.5 Summary

This chapter outlined the need for an automated Internal Examination Management System and explored its feasibility from economic, technical, and behavioral perspectives. We gathered requirements through interviews, surveys, and analysis of existing manual processes, ensuring they align with user needs. The validated requirements form the foundation for the system's design and implementation, addressing inefficiencies in question paper management, seating arrangements, and coordination.

Chapter 3

System Design and Methodology

The design phase is where the project's vision takes shape, defining its direction, structure, and the steps needed to bring it to existence. This phase involves planning, brainstorming, and creating a detailed blueprint that guides the entire development process. It acts as a road-map ensuring the final product fulfills its intended purpose. The design process unfolds in two key stages, each with a distinct focus:

- *System Design:-* This stage sets the big picture, outlining the overall structure and components of the project.
- *Detailed Design:-* Here, the focus narrows down to the specifics, creating a comprehensive blueprint with all the intricate details needed for successful execution.

3.1 System Design

- **Module 1: Front-End**
  - Handles user interaction and provides interfaces for all user roles, including Exam Coordinator (EC), Assistant Exam Coordinator (AEC), Module Coordinator (MC), Faculty, and HOD.
  - Ensures a user-friendly and responsive interface for seamless operation.
- **Module 2: Connection Module**
  - Acts as a middle-ware layer that connects the front-end with the database.
  - Manages business logic, request handling, and secure data exchange.
- **Module 3: Database**
  - Centralized storage for managing user profiles, scrutiny details, question paper data, room details, and batch details.
  - Provides data consistency and security for all operations.

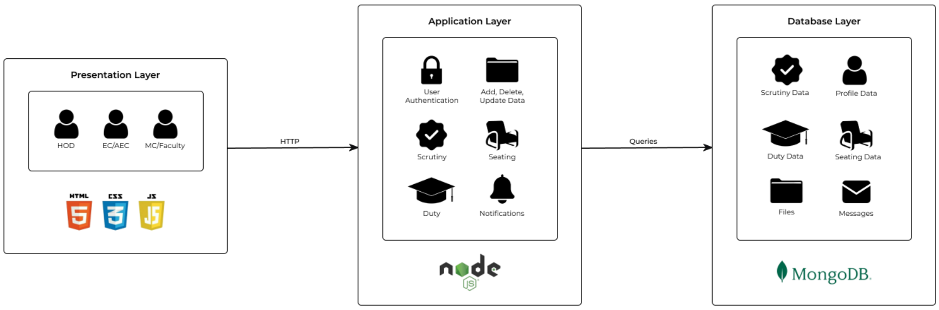

Figure 3.1: System Architecture Diagram

3.1.1 Module 1: Front-End

The front-end module of this system serves as the user interface, providing an intuitive and responsive platform for different user roles, such as the HOD, EC, AEC, Faculty, and MC. It ensures seamless interaction by offering role-specific pages and functionalities tailored to the needs of each user.

The key pages in the system are as follows:

1. **User Login, Forgot Password, and Change Password Pages** Handle authentication and account security.

2. **HOD Pages** Manage users, modules, and administrative tasks.

3. **EC/AEC Pages** Oversee exam-related tasks, including scrutiny allocation, subject and batch management, room assignments, seating arrangements, and duty allocation.

4. **Faculty/MC Pages** Facilitate question paper uploads, notifications, and messaging.

3.1.2 Module 2: Connection

The connection module manages interactions between the front-end and database, ensuring user requests are processed efficiently. It handles authentication, validation, session management, and data security using encryption and access controls. Acting as middleware, it optimizes performance, reduces latency, and enables scalability while supporting real-time data synchronization. Additionally, it ensures error handling, load balancing, and API management, maintaining system reliability and efficiency. By decoupling the front-end from direct database access, it enhances modularity, making maintenance and future updates easier.

3.1.3 Module 3: Database

The database module efficiently stores, manages, and retrieves data for internal exam management, including user profiles, subjects, rooms, batches, scrutiny allocations, and faculty activities. It ensures high availability and reliability for seamless scheduling, seating, and notifications. Security measures like encryption and role-based access control protect sensitive data, while its scalable design supports growing user demands and future expansions.

3.2 Detailed Design

In the detailed design, the abstract concepts and specifications outlined in the architectural design are elaborated upon to create a comprehensive blueprint for the software's implementation. It focuses on the specific implementation details, database, algorithms, and interactions required to build the system effectively.

3.2.1 Module 1: Front-End

The front-end module is responsible for the user interface and interaction with the system. It is designed to be responsive, user-friendly, and role-specific, ensuring seamless navigation for HOD, EC, AEC, Faculty, and MC. The key features are:

- Role-Based Dashboards: Displays relevant actions and information for each user.
- Authentication & Security: Ensures secure login, password management, and access control.
- Exam Management Tools: Includes seating arrangement, question paper upload, and scrutiny allocation features.
- Notifications & Messaging: Facilitates communication among users.

The front-end is built using HTML, CSS, JavaScript, and modern frameworks, ensuring efficiency, accessibility, and a smooth user experience.

3.2.2 Module 2: Connection

3.2.2.1 User Authentication:

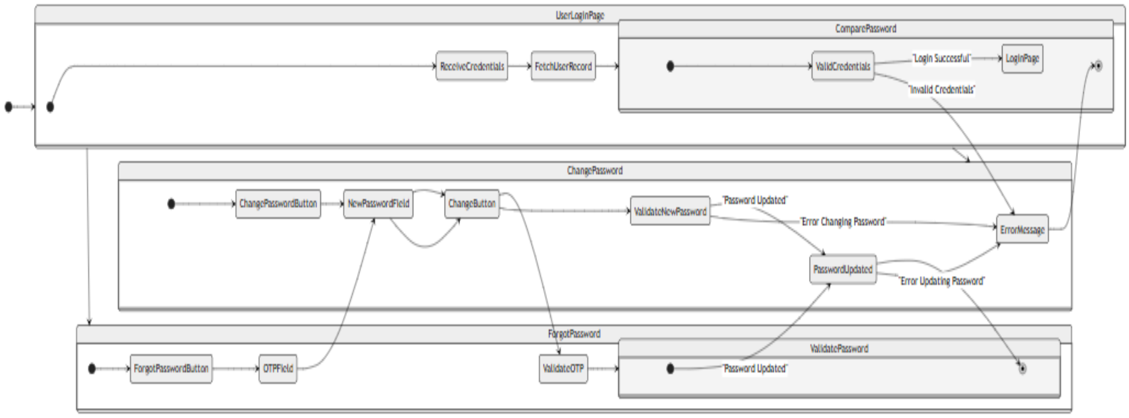

Figure 3.2.2.2: State Chart Diagram of User Authentication

Algorithm 1: User Login

```
Input: User details (Username and Password)
Output: Login to the respective page
Initial Screen: User Login Page
if inputs are valid then
    if username exists in the database then
        Compare the provided password with the password in the database;
        if password is correct then
            Login to the page;
        end
        else
            Display error message: "Invalid password";
        end
    end
    else
        Display error message: "Username not found";
    end
end
else
    Display error message: "Invalid input";
end
```

3.2.2.2 User Management:

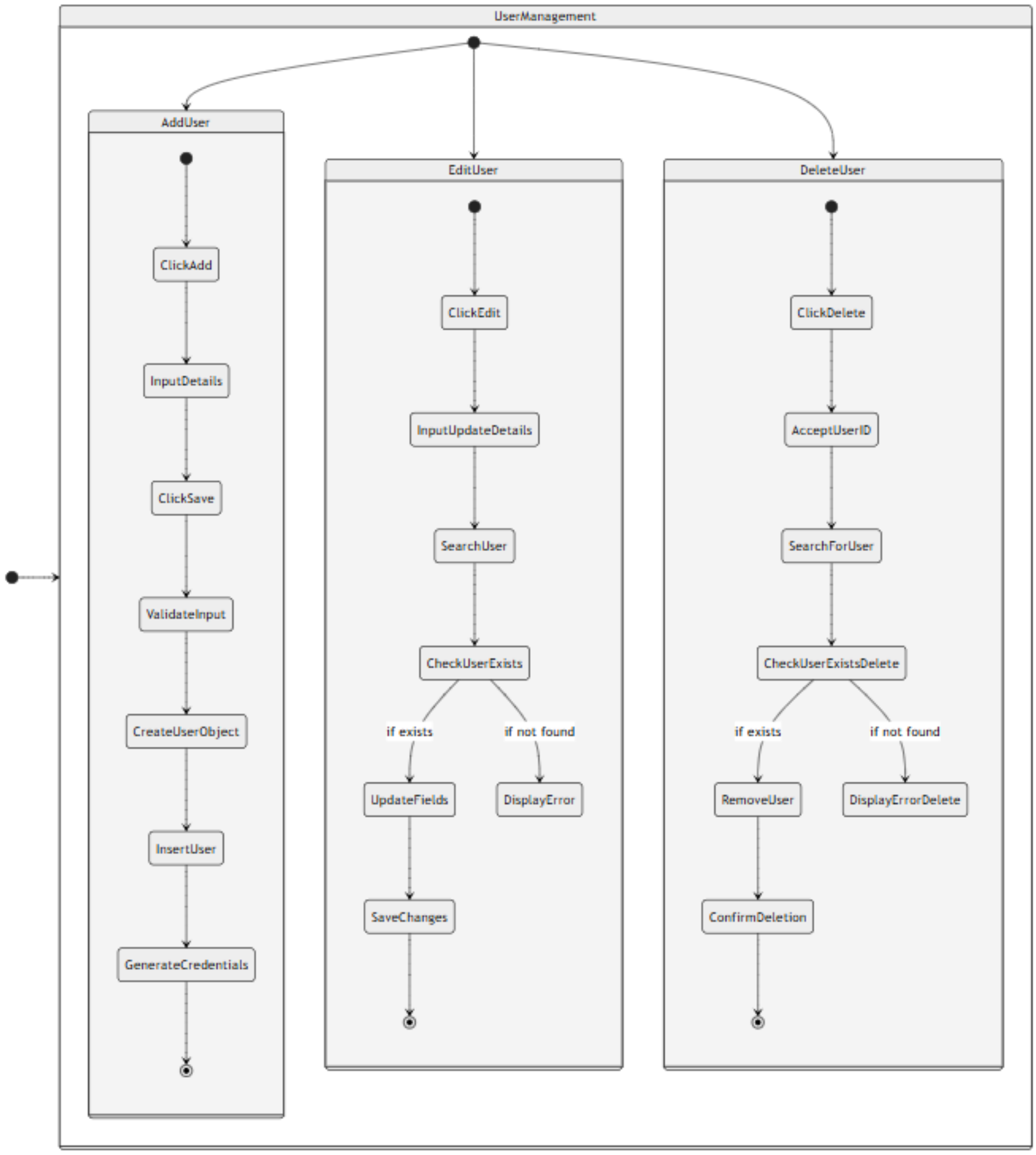

Figure 3.2.2.3: State Chart Diagram for User Management

Algorithm 2: Add User

Input: User details (Name, Phone Number, Email, Roles, Username) or a CSV file
Output: New user(s) added to the database

```
Initial Screen: User Management Page;
if User selects "Add New User" then
    Open the Add User Form;
    Accept input fields: Name, Phone Number, Email, Roles, Username;
    if inputs are valid then
        if username does not exist in the database then
            Store the user details in the database;
            Display success message: "User added successfully";
        end
        else
            Display error message: "Username already exists";
        end
    end
    else
        Display error message: "Invalid input";
    end
end
else if User selects "Download Template" then
    Generate and download an Excel file template;
end
else if User selects "Upload CSV" then
    Open file upload dialog;
    if file format is not .csv then
        Display error message: "Invalid file format. Please upload a .csv file";
    end
    else
        Upload file to backend;
        Read file contents and parse user details;
        foreach row in the file do
            Create an object for each user with the provided details;
        end
        Display success message: "Users uploaded successfully";
    end
end
Refresh User Management Page;
```

3.2.2.3 Module Management:

Algorithm 3: Add Module

Input: Module details (Name, Module Coordinator, Subjects, Faculties)
Output: New module added to the database

Initial Screen: Module Management Page;
Click on Add New Module;
Input module details: Name, Module Coordinator, Subjects, Faculties;
Click on the Save button;
Validate the input fields:;
    Ensure the module name is unique;
    Check that coordinator and faculties are valid users;
Create a new module object;
Insert the module object into the Module collection;

3.2.2.4 Subject Management:

Algorithm 4: Add Subject

Input: Subject details (Name, Code, Batch, Module Name, Faculty)
Output: New subject added to the database

Initial Screen: Subject Management Page;
Input subject details: Name, Code, Batch, Module Name, Faculty;
Validate the input fields:;
    Check that faculty corresponds to a valid user;
    Check that the batch exists in the database;
    Check that the module name exists in the database;
Create a new subject object;
Insert the subject object into the Subject collection;

3.2.2.5 Room Management:

Algorithm 5: Add Room

Input: Room details (Room No. and Capacity)
Output: New room added to the database

Initial Screen: Room Management Page;
Input room details: Room No. and Capacity;
Validate the input fields;
Create a new room object;
Insert the room object into the Room collection;

3.2.2.6 Batch Management:

Algorithm 6: Add Batch

Input: Batch details (Batch name and Strength)

Output: New batch added to the database

1 **Initial Screen:** Batch Management Page;
2 Input Batch details: Batch name and Strength;
3 Validate the input fields;
4 Create a new batch object;
5 Insert the batch object into the Batch collection;

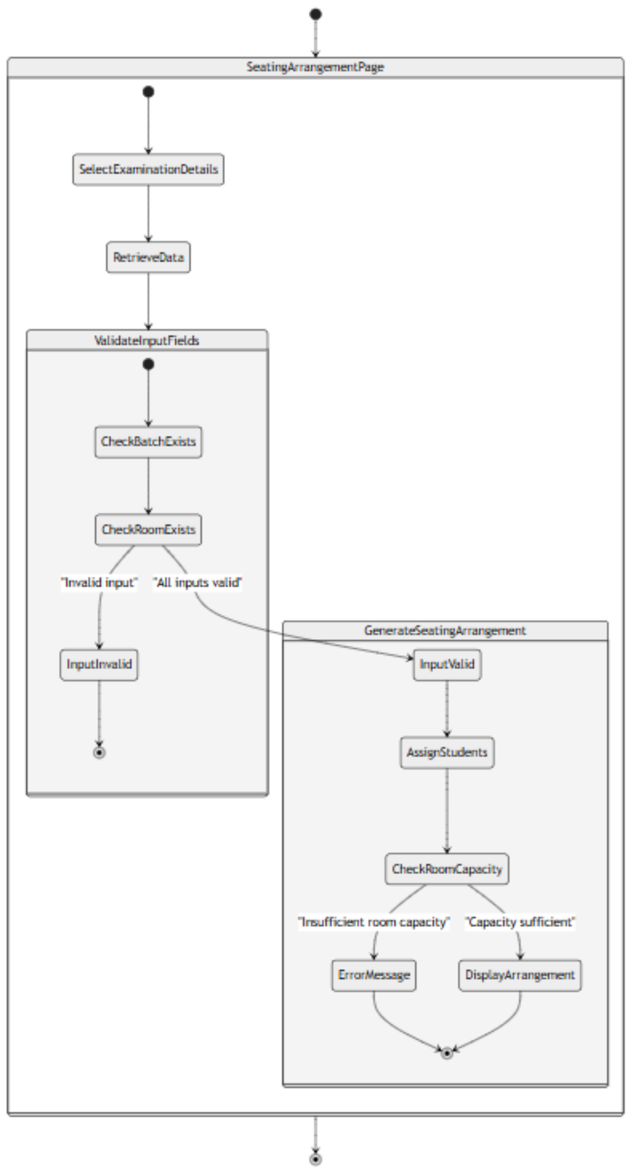

Figure 3.2.2.4: State Chart Diagram for Seating Arrangement

3.2.2.7 Seating Arrangement:

Algorithm 7: Seating Arrangement

```
Input: Selected Batches, Selected Rooms, Exam Date, Exam Name
Output: Seating arrangement for students
Step 1: Fetch Batch and Room Details
Retrieve batch details from the database based on selected batches;
Retrieve room details from the database based on selected rooms;
Step 2: Capacity Check
Calculate total students from selected batches;
Calculate total seating capacity from selected rooms;
if total capacity < total students then
    Display error message: "Insufficient room capacity to seat all students";
    Terminate algorithm;
end
Step 3: Generate Seating Arrangement
Arrange the rooms in a random order;
Initialize seating index as 1;
while students remain unseated do
    Assign students from the first batch to the current index;
    Increment the index based on the number of batches;
    Assign students from the next batch to the new index;
end
Repeat until all students are seated;
```

3.2.2.8 Scrutiny Allocation:

Algorithm 8: Faculty Submission

Input: Subject, Class, Trial Question Paper

Output: Final Question Paper Submission

```
Initial screen: scrutiny allocation page;
Faculty selects subject and class;
Faculty uploads the trial version of the question paper;
Trial version is sent to the inbox of the Module Coordinator;
Module Coordinator Assigns Scrutiny:
Module Coordinator assigns a faculty for scrutiny;
Question paper is sent to the Scrutiny Section in the inbox of the assigned faculty;
Scrutiny Process:
Scrutinizing faculty clicks the "Scrutiny" button to open a scrutiny form;
After filling out and submitting the form, a final submit button appears;
Upon final submission, the question paper along with the scrutiny form moves to the
   Verify Section in the Module Coordinator's inbox;
Verification by Module Coordinator:
Module Coordinator can add extra descriptions in a description box;
After reviewing, Module Coordinator clicks the "Verified" button;
Scrutiny form and description are sent to the faculty's inbox;
Final Question Paper Submission:
Faculty uploads the final question paper by selecting "Model Final";
Final question paper reaches the inbox of the Exam Coordinator;
```

3.2.3 Module 3: Database

The backend module processes user requests, manages authentication, and handles core exam-related operations like question paper scrutiny, seating arrangements, and duty allocation. It ensures secure access control, efficient database communication, and system reliability through encryption and error handling. Built with Node.js, it enables scalability, performance optimization, and seamless interaction between the front-end and database.

3.3 Summary

The proposed system for Internal Exam Management System consists of different modules for the UI, database, and connectivity, making sure they work together smoothly. In the detailed design phase, the organization of each functionality was described using state chart diagrams and algorithms.

Chapter 4

Implementation

The Implementation phase involves transforming the system design into a fully functional application. It includes developing, testing, and integrating different modules to ensure seamless operation. The system is implemented using modern web technologies to provide an efficient, scalable, and secure platform for internal examination management.

4.1 Tools Used

Implementing all the necessary features in a project of this magnitude demands the use of numerous tools. Here is the list of tools utilized for the development of this project.

4.1.1 Figma

Figma, a robust collaborative design platform, streamlined UI development for our website. Its real-time cloud-based collaboration and design-sharing capabilities ensured alignment with our vision. Its design-sharing capabilities facilitated a smooth transition from design to implementation, ensuring alignment with our vision and user needs. Figma's responsive design tools and integrations with platforms like GitHub enhanced collaboration.

4.1.2 HTML, CSS, JavaScript

These technologies power the front-end, creating a responsive and interactive user interface. JavaScript enhances functionality, while CSS ensures an intuitive design.

4.1.3 Node.js

Node.js is used for backend development, enabling efficient request handling, authentication, and data processing. Its asynchronous, event-driven architecture ensures scalability and performance.

4.1.4 MongoDB

MongoDB, a NoSQL database, serves as the database, efficiently storing and managing user data, exam details, and seating arrangements. Its document-based model organizes data into collections, facilitating easy management and flexible querying. Additionally, MongoDB securely stores authentication details, and its scalability and fault tolerance ensure a seamless user experience.

4.1.5 Express.js

Express.js is a lightweight web framework used with Node.js to streamline API development, request handling, and middleware integration.

4.1.6 Session Tokens

Session tokens are used for user authentication and session management, ensuring secure access control for different user roles.

4.1.7 SMTP (Simple Mail Transfer Protocol)

SMTP is used for email notifications, allowing the system to send alerts and important updates to users regarding exam schedules, scrutiny assignments, and duty allocations.

4.1.8 Visual Studio Code

Visual Studio Code(VSCode) was essential for our web development. This open-source code editor with versatile extensions, integrated Git support, and powerful debugging tools streamlined our coding and collaboration. Its intelligent code completion and customizable themes boosted productivity, making it very useful for our development process.

4.1.9 Postman

Postman was crucial in our web development. This API testing tool enabled seamless interaction with different APIs, ensuring effective integration between various components. With Postman, we validated routes and functionalities, guaranteeing smooth data exchange between the frontend and backend.

4.1.10 GitHub

GitHub is used as a collaboration platform, enabling version control, team coordination, and efficient code management throughout the development process.

4.2 Module Implementation

In this section of our project report, the implementation of each module is explained in detail.

4.2.1 Front-End Implementation

Front-end Module We developed our frontend module using HTML, CSS, and JavaScript to create a responsive and user-friendly interface. Our approach focused on clean, maintainable code with structured styling and dynamic interactivity.

To enhance the user experience, we leveraged JavaScript for dynamic content updates and seamless interactions. CSS was used to ensure a visually appealing design with responsive layouts, while structured HTML provided a strong foundation for accessibility.

Additionally, we incorporated various libraries such as Font Awesome for icons, and Fetch API for efficient API communication, ensuring a smooth and interactive user experience.

4.2.2 Connectivity Module

The Connectivity module is a critical component of any web application, designed to facilitate seamless communication and data exchange between the Front-end module and Database module of the system. This module ensures that users can interact with the website's features efficiently. It leverages various technologies and frameworks to handle HTTP requests, manage user sessions, and maintain data consistency across the application.

4.2.2.1 Implementation of features in detail

A detailed description of the implementation of each feature in DEPARTMENT STOCK MANAGEMENT SYSTEM is included in this section.

1. **User Authentication**

   The **User Authentication** feature ensures secure access to the system based on role-based authentication. It includes user login,and session management. The authentication process is implemented using javascript for the frontend and Node.js with Express.js for backend API communication.

   Login Process:
   The login page is built using HTML,CSS and javascript The *handleSubmit* function is triggered upon clicking the login button. It validates inputs, then makes an HTTP POST request to the backend authentication endpoint using *axios.post*. The backend verifies credentials and, upon successful authentication, returns a sessoin token.

The frontend stores this token in *sessionStorage.setItem* and navigates users to their respective dashboards using *navigate*. **Key Methods Used:**

- *axios.post* – To send authentication requests to the backend.
- *sessionStorage.setItem* – To store the authentication token securely.
- *navigate* – To redirect users based on their role.

2. **Scrutiny Functionality**

   - **Trial Version Question Paper Uploading By Faculty:**
     - Faculty selects mode as trial, batch, and subject, then uploads the question paper.
     - The form is built using HTML forms, and upon submission, fields are validated.
     - If validation passes, data is sent to the backend via an `axios.post` request and stored in the trial collection.
     - If validation fails, an error message is displayed.
     - On successful submission, the question paper is stored in the database and displayed in the inbox of the respective module coordinator.
   - **Scrutiny Allocation To Faculty By Module Coordinator:**
     - The question paper appears in the module coordinator's inbox along with a faculty selection dropdown and an allocate button.
     - The dropdown displays the names of faculty members.
     - When the allocate button is clicked, the selected faculty is validated.
     - If no faculty is selected, an error message is shown.
     - Upon validation, the allocated faculty's name is sent to the backend via `fetch`, and the corresponding trial collection record is updated.
     - The question paper is then displayed in the inbox of the allocated faculty.
   - **Question Paper Scrutiny By Allocated Faculty:**
     - Once allocated, the question paper appears in the faculty's inbox.
     - The faculty can click the "scrutiny" button to open a digital scrutiny form.
     - After filling out the form and submitting, the data is displayed in a message box.
     - Clicking the "done" button triggers field validation.
     - If validation fails, an error message is shown.
     - Upon successful validation, data is sent to the backend via `fetch`, where the corresponding trial record is updated to "scrutinized."

    - The scrutiny form is stored in the database and the scrutinized question paper is sent to the inbox of the faculty who uploaded the trial version.

- **Final Version Question Paper Uploading By Faculty:**
    - After reviewing the scrutiny form, the faculty creates the final version of the question paper and uploads it with the subject and batch.
    - Fields are validated during the upload process.
    - If validation fails, an error message is shown.
    - If validation passes, the data is sent to the backend via `fetch`.
    - A post API adds the new field to the trial module and the "Final" mode question paper is sent to the inbox of the EC.

Key Methods Used:

- `axios.post` – To send stock forwarding requests to the backend.
- `axios.get` – To fetch available stock and premises from the database.
- *HTTP methods (GET, POST, PUT, DELETE)* – For CRUD operations across the different profiles.
- `sessiontoken` – To authenticate the user.

3. **Automated Seating Arrangement**

The **Automated Seating Arrangement** feature automates the allocation of students to examination rooms, taking into account room capacities and class sizes.

Seating Arrangement Process:
The process begins when the EC or Assistant Examination Coordinator (AEC) initiates the seating arrangement via the front-end. A request is sent to the backend to retrieve class and room data. The backend processes this data and returns an optimized seating arrangement list. The generated seating plan is displayed to the user for review.

Key Methods Used:

- *axios.post* – To initiate the seating arrangement process.
- *axios.get* – To retrieve the generated seating plan.

4. **Invigilation Duty Allocation**

The **Invigilation Duty Allocation** feature assigns invigilation duties to ensure that each examination room is properly monitored. The process is straightforward: the EC/AEC manually selects the faculties and the rooms, and the backend generates a

PDF containing the invigilation assignments with faculties randomly assigned to the selected rooms.

Invigilation Duty Allocation Process:

- The EC/AEC triggers the invigilation duty allocation process via the front-end interface.
- The user selects the faculties and the corresponding examination rooms.
- The backend processes these inputs, randomly assigns the selected faculties to the rooms, and generates a PDF document containing the allocation details.
- The generated PDF is automatically downloaded for the user.

Key Methods Used:

- *axios.post* – For initiating the duty allocation process.
- *API endpoints* – For fetching and updating the assignment data.
- *Node.js libraries (e.g., PDFKit)* – For generating the PDF document with the invigilation assignments.

5. **Profile Management**

The **Profile Management** feature provides comprehensive CRUD operations for managing key system profiles. It covers five distinct profiles:

Room Management:

- **Data:** Room code and capacity.
- **Management Role:** EC/AEC handles room data entry and updates.

Batch Management:

- **Data:** Batch name and strength.
- **Management Role:** EC/AEC handles creation and modification of batches.

Subject Management:

- **Data:** Subject name, code, associated batches, and faculties.
- **Management Role:** EC/AEC is responsible for managing subject details.

Module Management:

- **Data:** Module name, module coordinator, faculties (sourced from User Management), and subjects (from Subject Management).
- **Management Role:** Managed by the HOD.

User Management:

- **Data:** Username, name, email, phone, and roles.
- **Additional Detail:** The password is auto-assigned a default value upon creation.
- **Management Role:** Handled by the HOD.

Workflow:

- Authorized users (EC/AEC for Room, Batch, and Subject Management; HOD for Module and User Management) interact with the front-end interface to perform CRUD operations.
- Each operation (Create, Read, Update, Delete) triggers an HTTP request (GET, POST, PUT, or DELETE) to the corresponding backend API endpoint.
- The database is updated accordingly, and the system ensures data consistency through real-time synchronization.

Key Methods Used:

- *HTTP methods (GET, POST, PUT, DELETE)* – For CRUD operations across the different profiles.
- *API endpoints* – Dedicated to each management area.

4.2.3 Database Module

Our web application's MongoDB database is divided into the following collections. The content of each collection is described below.

(a) **User Collection**:

Stores user details and authentication credentials. Each user is identified by a unique `U_id` and contains fields such as `U_name` for the user's name, `U_phone` for contact details, `U_email` for the email address, `U_role` to define the user's role (e.g., FC, MC, AEC, etc.), and `U_password` for secure authentication. This collection plays a crucial role in managing user authentication and authorization.

(b) **Room Collection**:

Stores information about rooms for seating arrangement automation. Each room is identified by its `R_code` and includes details such as `R_capacity`, which specifies the seating capacity. This collection helps efficiently organize seating arrangements within the institution.

(c) **Subject Collection**:

This collection contains subject-related details, primarily used for the scrutiny system. It stores `S_name` for the subject's name, `S_code` as the subject code, `S_batch` to indicate the batch taking the subject, and `S_faculty` to track the faculty responsible for teaching the subject.

(d) **Notify Collection**:

This collection is used for storing messages sent through the notification functionality of the system. It includes `from` to indicate the sender's name, `to` for the recipient's name, `filename` for any attached files, `data` for storing attached PDFs in Base64 format, and `textmessage` to hold the message content sent along with the notification.

(e) **Module Collection**:

This collection stores module details, mainly used in the scrutiny functionality. It includes `moduleName` for the module's name, `moduleCoordinator` for the name of the coordinator, `subjects` listing the subjects associated with the module, `faculties` listing the faculty members involved, and `createdAt` to track the date of module creation.

(f) **Batch Collection**:

This collection stores batch details and is primarily used for seating arrangement automation. It contains `B_name` to specify the batch name and `B_strength` to denote the number of students in the batch.

(g) **XL Collection**:

This collection stores plain Excel files with predefined headings used for bulk user management. The files uploaded here facilitate adding multiple users at once, streamlining the user management process.

(h) **Trial Collection**:

This collection serves as the foundation of the scrutiny system, storing all details related to the scrutiny process of question papers. It includes `mode` to indicate the current stage of the scrutiny process (e.g., trial, final, verified, scrutinized), `subject` for the subject to which the question paper belongs, `batch` for the batch associated with the paper, `fileName` for the name of the uploaded question paper file, and `data` for storing the question paper in Base64 format. Additional fields include `date` for tracking the upload date, `from` to indicate the faculty member who uploaded the question paper, `allocate` for the faculty assigned to review it, `sfileName` for the name of the scrutiny form uploaded by the reviewing faculty, `sdata` for the scrutiny form data stored in Base64 format, and `textmesg` for suggestions provided by the module coordinator after scrutiny.

4.3 Summary

During the implementation phase, the Internal Examination Management System was developed using modern web technologies to ensure security, scalability, and efficiency. The front-end backend and database were implemented with tools like HTML, CSS, and JavaScript, Node.js, MongoDB, and Express.js to provide a seamless user experience. Additionally, SMTP for email notifications and GitHub for collaboration enhanced system functionality and development efficiency. This chapter detailed the tools used and the module-wise implementation of the system.

Chapter 5

Testing

Software Testing[3,4] is a method to check whether the actual software product matches expected requirements and to ensure that the software product is Defect free. It involves the execution of software/system components using manual or automated tools to evaluate one or more properties of interest. The purpose of software testing is to identify errors, gaps or missing requirements in contrast to actual requirements.

5.1 Testing Strategies Used

In this section, we will discuss the testing strategies employed to ensure the reliability and functionality of our web-applcation. Testing played a crucial role in validating the site's performance and accuracy. To ensure the reliability and functionality of the Internal Examination Management System, multiple testing strategies were employed.

- Unit Testing Individual components, such as authentication, profile management, and question paper uploads, were tested to ensure correctness.
- Integration Testing Verified interactions between front-end, backend, and database, ensuring smooth data flow and API functionality.
- System Testing End-to-end testing of the complete system to confirm that all modules function as expected.
- Regression Testing Previously validated features were re-tested after updates to ensure no new issues were introduced.
- Security Testing Assessed vulnerabilities, encryption mechanisms, and access control to prevent unauthorized data access.
- UI Testing Evaluated user experience, responsiveness, and accessibility across different devices and browsers.

5.2 Testing Results

The above-mentioned testing strategies were implemented in our project. The following are the results obtained.

5.2.1 Results of Black Box Testing

Black box testing was conducted to evaluate the system's functionality from an end-user perspective. Testers performed operations such as login, user management, module management, subject management, room management, batch management, seating arrangement, duty allocation, scrutiny allocation and notification management. Errors identified during testing and their resolutions are as follows:

(a) *User Authentication*: Some invalid login attempts were not handled correctly, allowing unauthorized access in certain cases. This issue was resolved by strengthening validation checks and enforcing authentication policies.

(b) *Notification System*: Notifications regarding duty and scrutiny allocations were not reaching the respective users. The issue was fixed by modifying the notification service to properly identify and map recipients.

(c) *Seating Arrangement Conflicts*: Some students were assigned multiple seats due to incorrect seat allocation logic. This issue was resolved by refining seat assignment algorithm to prevent duplicate seating.

After resolving these issues, the system performed as expected across various modules.

5.2.2 Results of White Box Testing

White box testing focused on reviewing the source code, logic flow, and database interactions. A trial database was created with test data to simulate real-world scenarios. One significant issue encountered was:

(a) *API Connectivity Issue*: Axios requests were failing due to incorrect base URLs. Initially, localhost URLs were used, which prevented access across networks. The issue was resolved by configuring API endpoints with dynamic environment-based URLs.

(b) *Session Management Bug*: User sessions were not expiring correctly, allowing prolonged access beyond the intended session timeout. This issue was fixed by implementing proper session expiration checks and ensured automatic logout after inactivity.

Post-testing, database consistency checks confirmed that stock transactions were accurately recorded and retrieved.

5.2.3 Results of Integration Testing

The integration of different system modules was tested to ensure seamless interactions. Some issues encountered included:

(a) *Mismatch in Data Models*: Certain fields were missing when retrieving seating arrangement, duty allocation, and scrutiny details due to inconsistencies in schema definitions. This was fixed by aligning frontend API calls with the backend database schema to ensure data consistency.

(b) *Notification Delivery Failures*: Some notifications were not being sent due to missing recipient details. The notification system was updated to ensure accurate message delivery.

(c) *Data Synchronization Delays*: Updates to room allocations and batch management were not instantly reflected across all modules. This was resolved by improving Updates to room allocations and batch management were not instantly reflected across all modules.

Once these issues were resolved, the integrated system was re-tested and verified to function correctly.

5.3 Summary

In this chapter, we described the various testing strategies used to test the Internal Exam Management System and the results obtained on applying these tests on thr system. The complete complete details As part of testing, various strategies such as unit, integration, system, regression, and security testing were conducted to ensure the system's functionality, security, and performance. Key features, including authentication, question paper management, seating arrangements, and notifications, were thoroughly tested. The results confirmed that all critical test cases passed, validating the system's efficiency, reliability, and readiness for deployment.

Chapter 6

Results

The Internal Exam Management System was successfully implemented and the following results were obtained.

6.1 Output Screenshots

Given below are the screenshots of the results obtained after the implementation and testing of Internal Exam Management System.

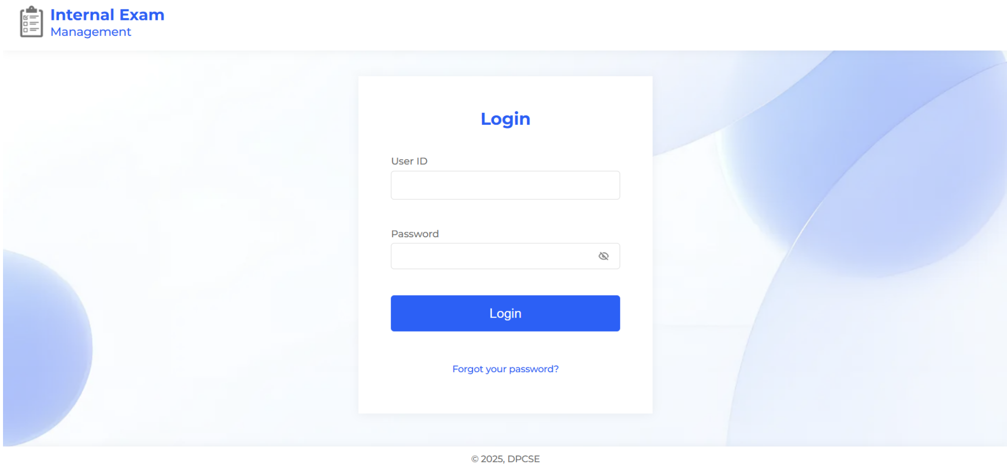

Figure 6.1.1: User Login Page

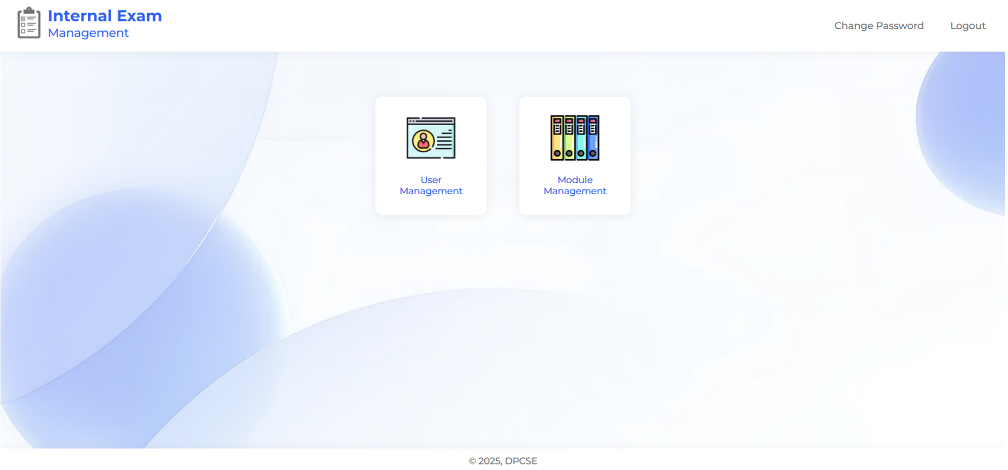

Figure 6.1.2: HOD Home Page

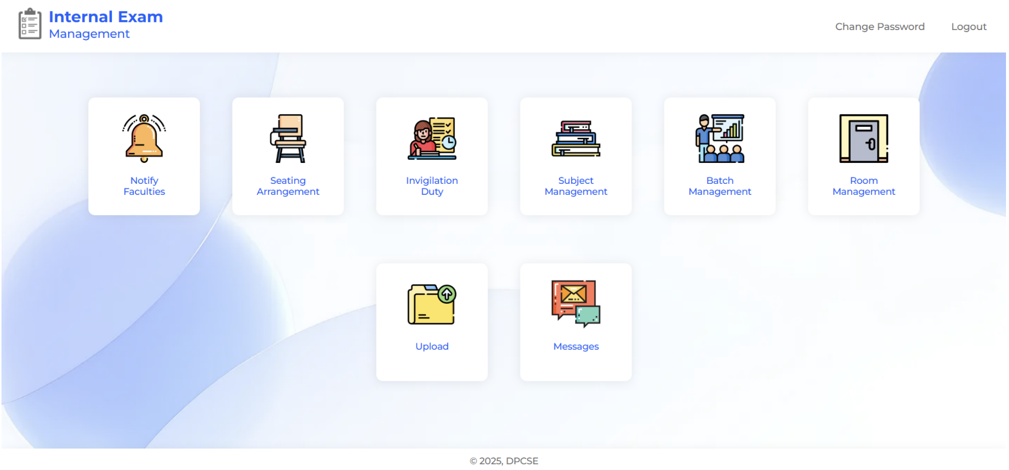

Figure 6.1.3: EC/AEC Home Page

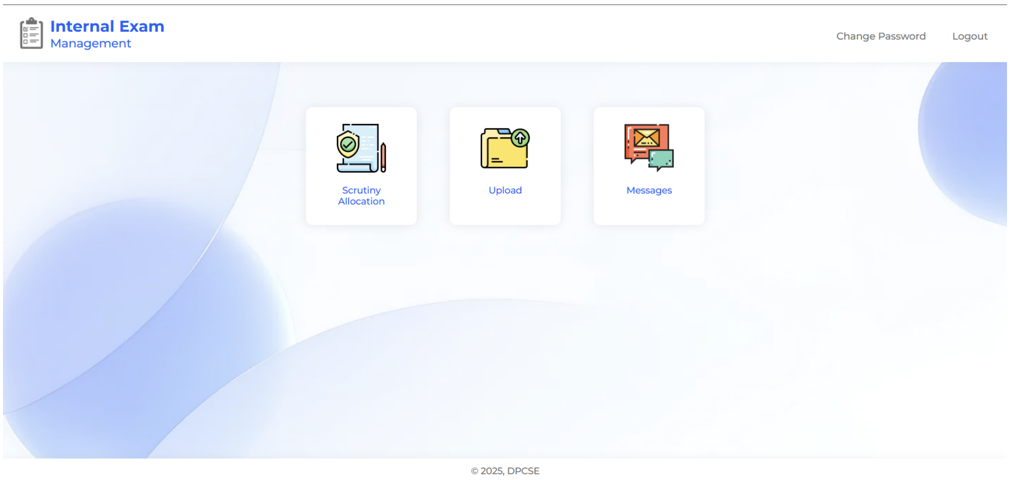

Figure 6.1.4: MC Home Page

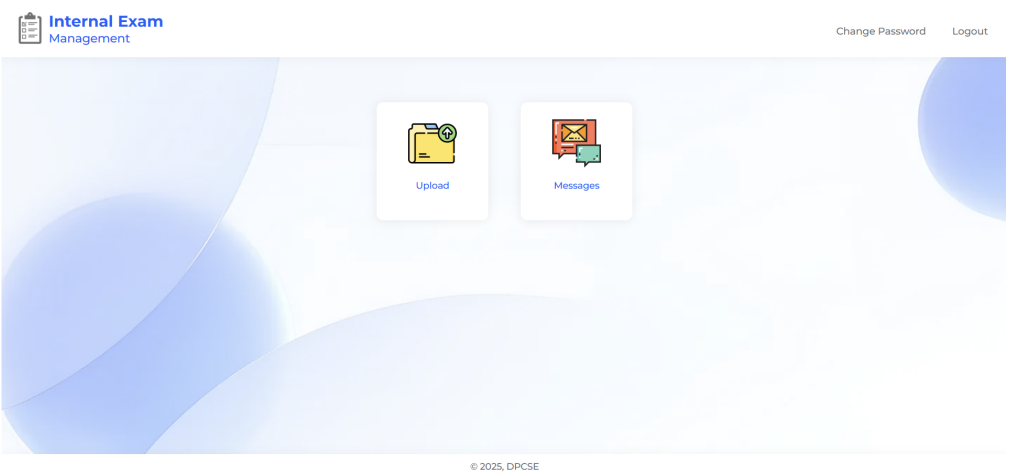

Figure 6.1.5: FC Home Page

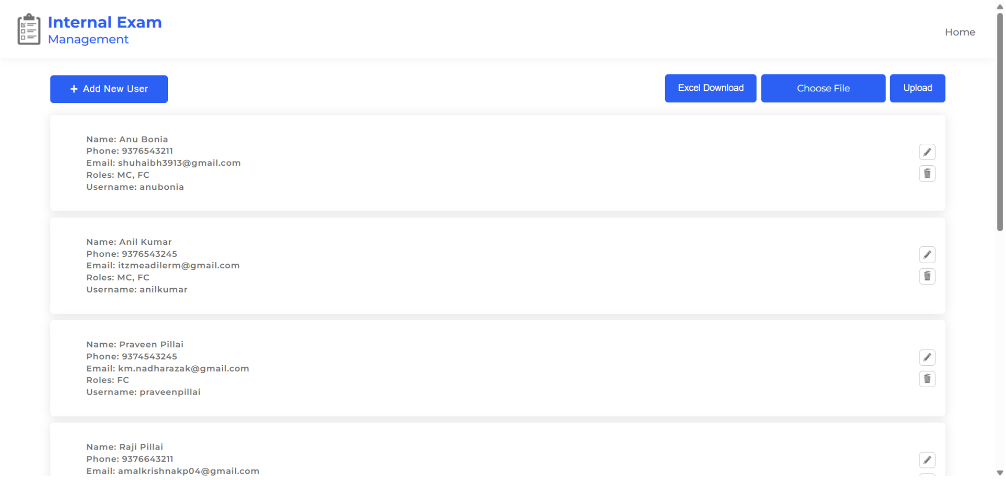

Figure 6.1.6: User Management Page

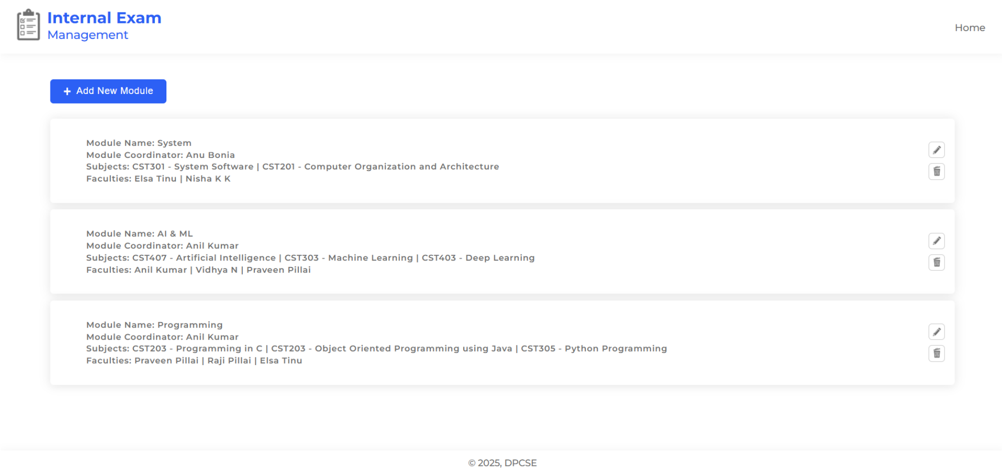

Figure 6.1.7: Module Management Page

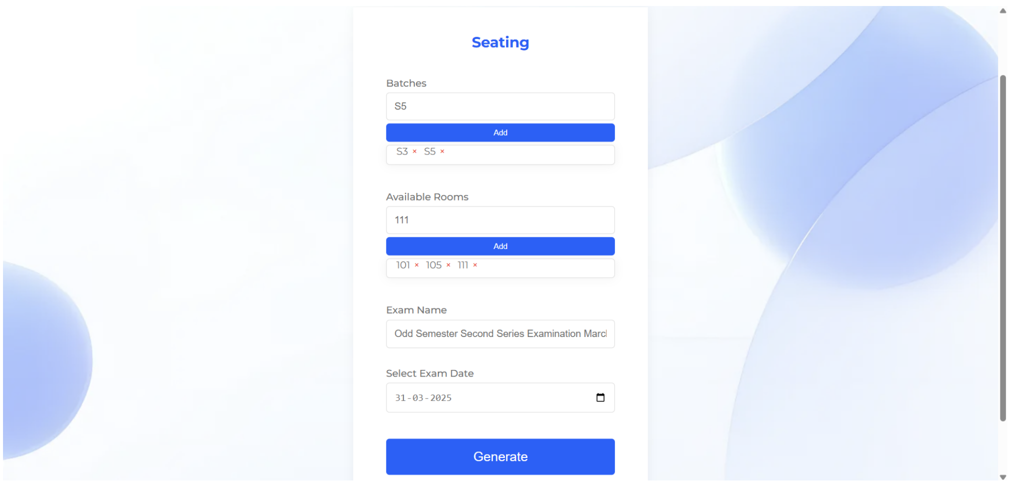

Figure 6.1.8: Seating Arrangement Page

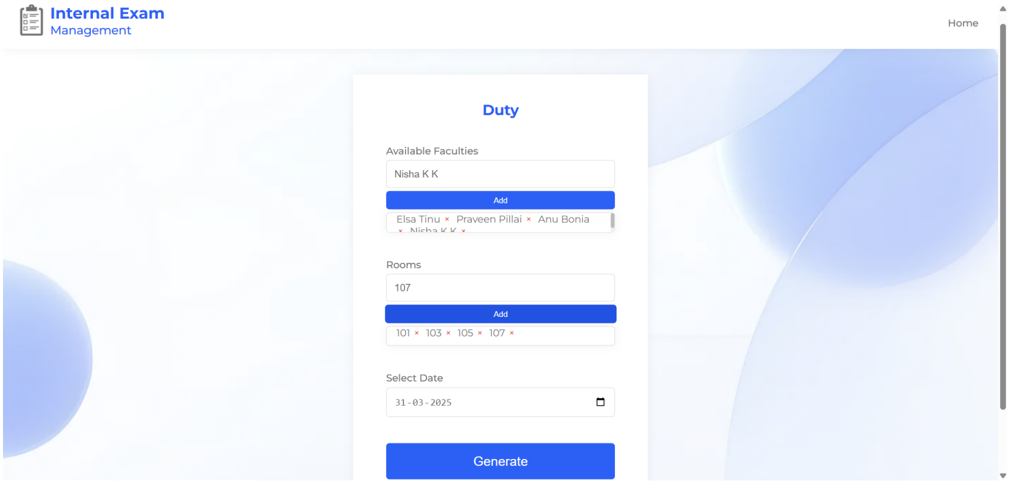

Figure 6.1.9: Duty Allocation Page

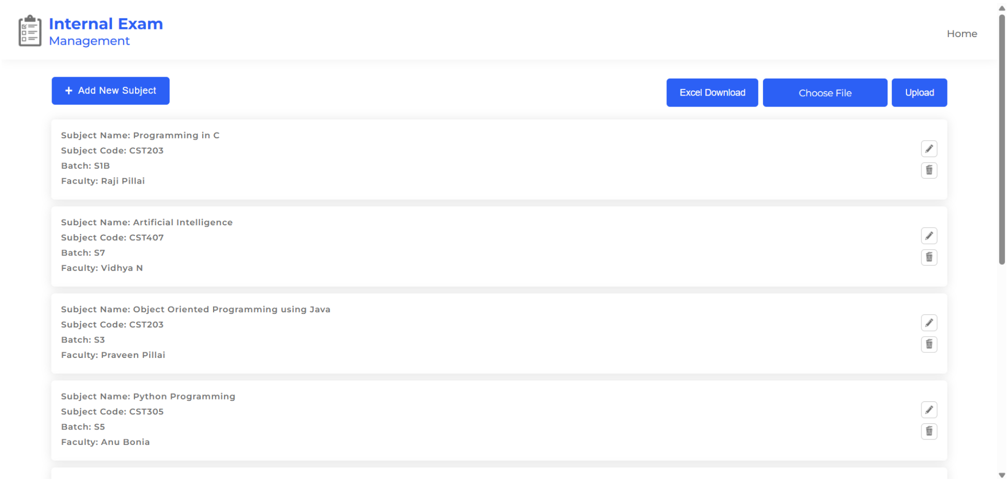

Figure 6.1.10: Subject Management Page

Figure 6.1.11: Batch Management Page

Figure 6.1.12: Room Management Page

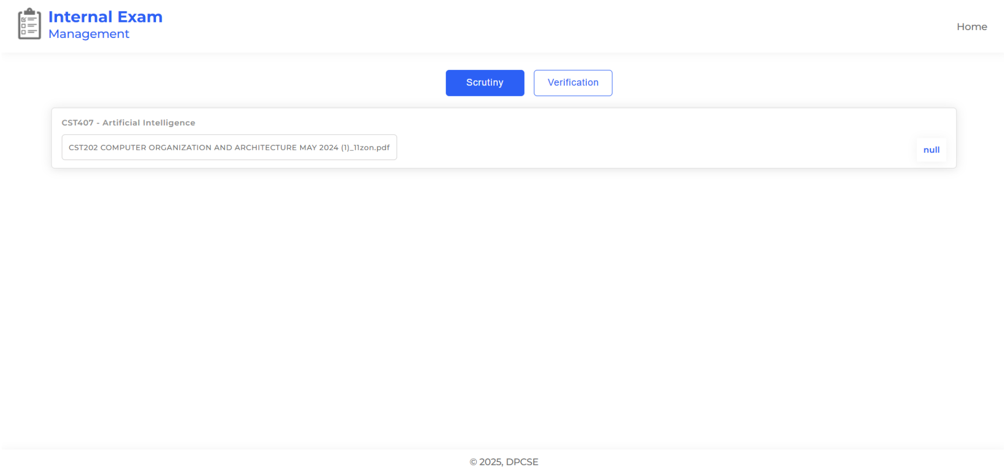

Figure 6.1.13: Scrutiny Allocation Page

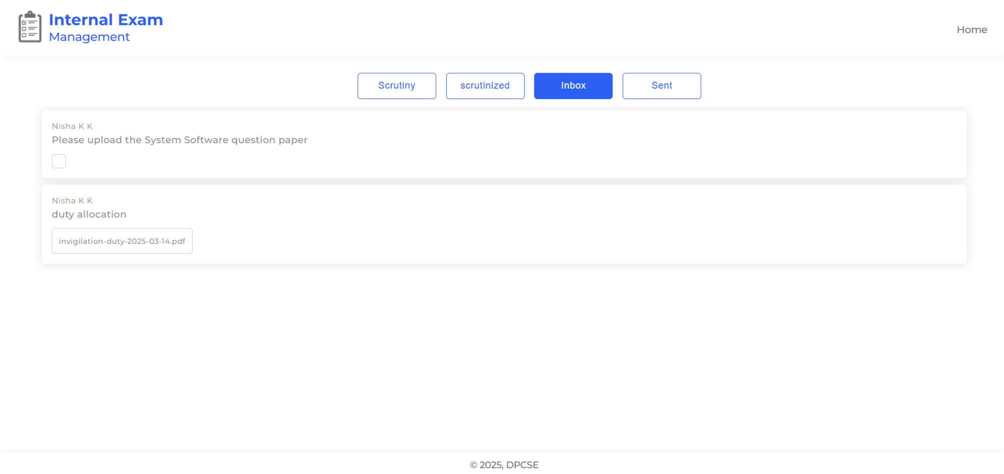

Figure 6.1.14: Messages Page

6.2 Result Analysis

Result Analysis aims to analyze the data collected, draw meaningful conclusions, and evaluate the extent to which the project objectives have been achieved. Through a systematic and comprehensive analysis of the results, the following inferences were made.

(a) The system successfully facilitates all core functionalities, including exam scheduling, student performance evaluation, automated result generation, and secure grade distribution.

(b) The system satisfies all non-functional requirements, ensuring:

- **Reliability:** The system ensures high availability, with minimal downtime during peak exam periods, and handles failures gracefully with error messages and recovery options.
- **Responsiveness:** It provides a smooth user experience, ensuring fast interactions such as uploading question papers and viewing schedules without delays.
- **Security:** Strong encryption protects sensitive data such as exam papers and seating arrangements. Role-based access control ensures appropriate user permissions, and secure authentication mechanisms, including multi-factor authentication for administrators, prevent unauthorized access.
- **Usability:** The system is designed to be intuitive and accessible across devices, providing a seamless experience for all users.
- **Testability and Maintainability:** A modular structure allows for efficient debugging and updates, with comprehensive logging and adherence to coding standards.
- **Compatibility:** It runs efficiently across multiple platforms, ensuring accessibility for all users.

6.3 Summary

In this chapter, the results of implementing the Internal Exam Management System were presented. by giving the sample output screens of different interfaces implemented. The fact of successful completion of the project was substantiated by a careful result analysis process whose complete details were also given in this chapter.

Chapter 7

Conclusion

7.1 Conclusion

In the preceding chapters, all the phases involved in the development of the Internal Exam Management System have been discussed in detail. This project was initiated to address the challenges associated with manual exam management systems, and enhance overall exam organization.

Before the development process began, a thorough requirement analysis was conducted. This involved studying existing exam management system and identifying the issues that needed to be resolved. The system requirements were finalized through detailed discussions with faculties, and a Software Requirements Specification (SRS) document was prepared to outline the functional and non-functional aspects of the system. These details were covered in Chapter 2.

Following this, the design phase was carried out, detailing the System Design and Detailed Design. The architecture was carefully planned using a modular approach, which included key users such as Head of Department, Exam Coordinator, Assistant Exam Cooridnator, Module Coordinators, and Faculties. The design followed best practices for maintainability and scalability, ensuring a structured and efficient workflow. These aspects were covered in Chapter 3.

Once the design phase was completed, the implementation phase began using modern web technologies such as HTML, CSS, MongoDB, Node.js, and Express.js. The development followed an iterative approach, ensuring incremental improvements and continuous integration of features. Upon completion of implementation, rigorous testing was performed, including unit testing, integration testing, and system testing. Issues found during testing were resolved to enhance system stability and performance. These details were discussed in Chapter 4.

The fully developed system was then deployed and evaluated based on functional and non-functional requirements. The results demonstrated that the system successfully met the objectives outlined in the requirement phase, providing an efficient exam management system. The findings and analysis of the developed system were presented in Chapter 6.

Through this structured approach, the Internal Exam Management System was successfully designed and implemented key functionalities such as automated seating arrangement, scrutiny allocation, subject management, room management, batch management and invigilation duty allocation. The project serves as a foundation for further enhancements, ensuring scalability and adaptability to future departmental needs.

7.2 Recommendations for Future Works

Despite its robust functionalities, the system can be further enhanced with additional features and optimizations. Some potential future improvements include:

- Mobile Application: A mobile-friendly version for easier access.
- AI-Based Optimization: Use AI algorithms for more efficient seating arrangements.
- Offline Mode Support: Allow certain features to function without internet connectivity.
- Student Access Module: Allowing students to view their seating arrangements and exam schedules.
- Enhanced Analytics Dashboard: Implementing graphical representation of data to assist in decision-making.
- History and Export: An export option to download faculty assignment history, including subject-wise coordinators and scrutiny faculty, within a specified date range.
- Automated Duty Allocation: Enhancing duty allocation to be more automated and fair.

These enhancements will further improve the system's usability, efficiency, and adaptability, ensuring it remains a valuable tool for internal exam management.

Appendix A

Software Requirement Specification

A.1 Introduction

A.1.1 Purpose

The purpose of this document is to provide a detailed description of the requirements of the project *Internal Examination Management System*. This document outlines the functional and non-functional requirements of the project, the constraints in which it operates, the deliverables, and the assumptions based on which it is built. It provides a clear understanding of the system's capabilities and constraints to its readers.

A.1.2 Document Conventions

HOD	Head of the Department
EC	Examination Coordinator
AEC	Assistant Examination Coordinator
MC	Module Coordinator

A.1.3 Intended Audience and Reading Suggestions

The intended audience of the document includes the team members, the project guide, the project coordinator. Team members rely on it for development, coding, testing, and maintenance, while the project guide utilizes the document to oversee project progress and to guide the team effectively. The project coordinator uses this document to evaluate whether the implementation is proper and analyze its functionalities.

A.1.4 Project Scope

The scope of the project include:

(a) **Need:**

- Lack of a proper automated system for handling Internal Examination in the Computer Science and Engineering Department of RIT Kottayam.

(b) **Deliverables:**

- A web application that enables to upload question papers, conduct question paper scrutiniy, set seating arrangement and allocate invigilation duty to the faculty.

(c) **Assumptions:**

- Users have the basic knowledge of handling web-based applications.
- Users have clear idea on examination management.

A.1.5 References

IEEE Std 830-1998, IEEE Recommended Practice for Software Requirements Specifications.

A.2 Overall Description

A.2.1 Product Perspective

The Internal Examination Management System is a Web application designed to integrate all the activities related to internal examination management.

A.2.2 Product Features

- Examination Schedule Management
- Faculty Assignation
- Question Paper Submission and Scrutiny
- Automated Seating Arrangements
- User-Friendly Interface

A.2.3 User Classes and Characteristics

- HOD
- Examination Coordinator

- Assistant Examination Coordinator
- Module Coordinators
- Teachers

A.2.4 Operating Environment

The Hardware and Software required for this project are mentioned in section 4.2 and section 4.3.

A.2.5 Design and Implementation Constraints

The design of the system must adhere to standard design conventions and programming practices to ensure consistency, scalability, and maintainability. This includes following platform-specific design guidelines, using modular code, and ensuring seamless integration with existing systems. Integration with Email gateways for notifications must also be considered, as limitations may arise. Frequent and effective communication with faculties is essential throughout the project to gather requirements, clarify doubts, and ensure the system meets their needs.

A.2.6 Assumptions and Dependencies

The project has the following assumptions:

- Users, including faculty and coordinators, are familiar with basic digital tools and can navigate a web-based interface.
- The system will be used primarily by authorized personnel (faculty, coordinators, and students) with role-based access control in place.
- Regular system updates and maintenance will be conducted to ensure security, functionality, and performance.

The project has the following dependencies:

- The performance of the system depends on the scalability and capacity of the underlying database and server infrastructure.
- The project depends on third-party services, such as email gateways, for notifications, which may have limitations.

A.3 System Features

The following are the features of this project:

- User Authentication
- Profile Management
- Question Paper Upload and Scrutiny
- Automated Seating Arrangement
- Automated Invigilation Duty Allocation

A.3.1 User Authentication

A.3.1.1 Description and Priority

This feature ensures secure access to the system based on user roles, Each valid user is suppose to be authenticated by the system so as to have access to the system. This feature enables to provide different interfaces and functionalities for various users, depending on their roles. This in turn ensures accountability and smooth workflow management.
Priority: High

A.3.1.2 Stimulus/Response Sequences

Stimulus:-

- User credentials.

Response:-

- Successful login if the user is an authenticated user. An Error message if the user is not an authenticated user.

A.3.2 Profile Management

A.3.2.1 Description and Priority

This feature manages

- The storage of the details of faculty in the department

- The storage of the details of various modules covering all the subjects within the academic curriculum.
- The storage of the details of module Committee Coordinators, Module Committee Members, Examination Coordinator, and Assistant Examination Coordinator.
- The storage of the details of each class such as the number of students.
- The storage of the details of each classroom such as the number of seats available.
- The storage of the details of each subjects and it's respective faculties.
- the examination timetable and its sharing to all faculty.

Priority:-High

A.3.2.2 Stimulus/Response Sequences

Stimulus:-

- HOD/EC should edit the details whenever needed.

Response:-

- profile details are updated.

A.3.2.3 Functional Requirements

(a) The HOD must be able to insert, delete, and update the details of the faculty in the department.

(b) The HOD must be able to add a new module and remove an existing module.

(c) The HOD must be able to update the details of the MC's, EC and AEC as needed.

(d) The EC and AEC must be able to enter and update the details of each subjects and it's respective faculties.

(e) The EC and AEC must be able to enter and update the details of each class such as the students of each class.

(f) The EC and AEC must be able to enter and update the details of each classroom such as the number of seats available.

(g) The EC and AEC must be able to share timetable to all faculties.

A.3.3 Question Paper Upload and Scrutiny

A.3.3.1 Description and Priority

This feature enables

- uploading of the trial version of the question paper(question paper before scrutiny) and the finalized question paper(question paper after scrutiny).
- Question paper scrutiny as per the existing rules in the department.

Priority:-High

A.3.3.2 Stimulus/Response Sequences

Stimulus:-

- Notification of the EC to the faculty for uploading the trial question paper

Response:-

- Scrutinized question papers are uploaded to the system.

A.3.3.3 Functional Requirements

(a) The EC must be able to notify the respective faculties for uploading the trial version of the question paper.

(b) The faculty must be able to upload the trial and final version of the question paper and notify the EC.

(c) The EC must be able to assign faculties for scrutiny and notify them

(d) The question papers must be visible for faculties and they must have a standard scrutiny form for marking their remarks and notify the MC.

(e) The scrutiny form must be visible to MC and they must can add additional remarks and notify the respective faculty.

A.3.4 Automated Seating Arrangement

A.3.4.1 Description and Priority

This feature automates the allocation of students to examination rooms based on room capacities and class sizes. ***Priority:-High***

A.3.4.2 Stimulus/Response Sequences

Stimulus:-

- Invocation by EC/AEC for Automated seating arrangement.

Response:-

- A list containing the invigilation duties for each faculty.

A.3.4.3 Functional Requirements

(a) The EC/AEC must be able to select the classes available and the batches that have the examination.

(b) The automated seating arrangement must be received.

A.3.5 Invigilation Duty Allocation

A.3.5.1 Description and Priority

This feature aids the allocation of faculties to examination rooms.

Priority:-High

A.3.5.2 Stimulus/Response Sequences

Stimulus:-

- Invocation by EC/AEC for invigilation duty allocation.

Response:-

- A list containing the invigilation duties for each faculty.

A.3.5.3 Functional Requirements

(a) The EC/AEC must be able to assign faculties to examination rooms and the faculties must be notified.

A.4 External Interface Requirements

A.4.1 User Interfaces

- **Login Page**: Allows users to securely log in to the system using their credentials and access their respective roles and functionalities.
- **Seating Arrangement Page**: Enables users to create, view, and modify seating arrangements for exams, ensuring optimal utilization of available rooms.
- **Duty Allocation Page**: Provides a platform to assign examination duties to faculty members, ensuring fairness and efficiency based on availability and requirements.
- **Question Paper Management Page**: Facilitates the upload, review, and approval of question papers through a streamlined workflow involving multiple coordinators.
- **Profile Page**: Allows the creation, update, and management of profiles for staff, subjects, classes, rooms, and roles to maintain organized and accurate data.

A.4.2 Hardware Interfaces

- **Processor:** Intel Core i3 or equivalent for client systems; Intel Core i5 or above for the web server.
- **RAM:** 4 GB or more for client systems; minimum 8 GB for the web server to handle moderate workloads efficiently.
- **Internet Connection:** A stable broadband connection is required for smooth access to the system.

A.4.3 Software Interfaces

- Operating system: Any modern operating system including Windows 7 or above, Mac OS, and Linux distributions can be used for both server and client machines.
- **Browser Support:** The client machine needs to have access to any of the modern web browsers like Google Chrome, Mozilla Firefox, or Microsoft Edge.
- **Backend Support:** The server size machine should be able to run a node js server.

A.4.4 Communications Interfaces

- Email Notifications

A.5 Other Nonfunctional Requirements

A.5.1 Performance Requirements

- Reliability: The system must ensure high availability, with minimal downtime during peak exam periods, and must be able to handle system failures gracefully, providing error messages and recovery options without data loss.
- Responsiveness: The system should provide a smooth user experience with fast and efficient interactions, ensuring that actions such as uploading question papers and viewing schedules are completed without significant delays or disruptions.

A.5.2 Security Requirements

- Data Protection: The system must implement strong encryption protocols to protect sensitive data, such as exam papers, and seating arrangements, both during transmission and at rest.
- Access Control: The system should enforce role-based access control to ensure that users can only access the features and data relevant to their assigned roles.
- Authentication and Authorization: The system must support secure login mechanisms, including multi-factor authentication for administrators and coordinators, to prevent unauthorized access.

A.5.3 Software Quality Attributes

- Usability: The system should have an intuitive, easy-to-navigate interface. It must be responsive and accessible across devices to ensure a smooth user experience.
- Testability and Maintainability: The system should be modular for easy testing and debugging, with comprehensive logging for tracking issues. It should follow standard coding practices, be well-documented, and support efficient updates and maintenance.

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